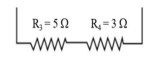
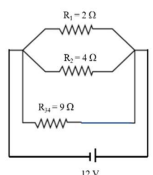
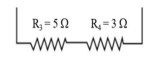
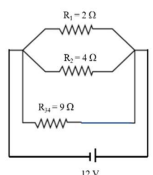
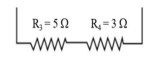
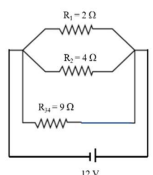
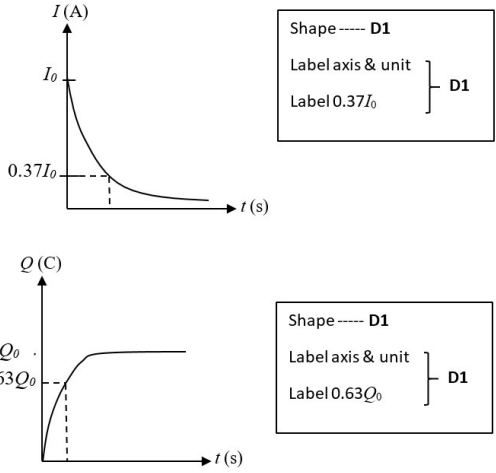
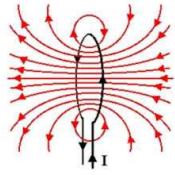


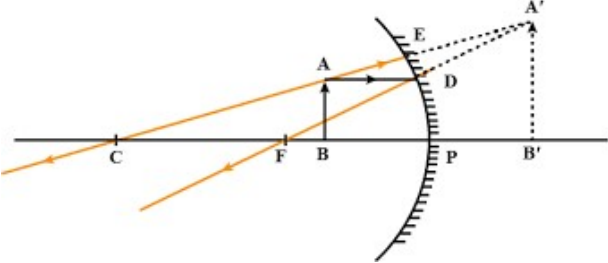
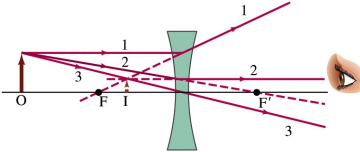
ANSWER SCHEME DAQ DP024 SESSION 2022/2023 SET A

1. a) i.	$v = \frac{dy}{dt} = A \omega \cos(\omega t + kx)$ $v = 6(16\pi) \cos(16\pi t + 0.8x)$ $v = 96\pi \cos((16\pi(0.1) + 0.8(4)))$ $v = -109.78 \text{ cm s}^{-1}$	G1 JU1
a) ii.	$\lambda = \frac{2\pi}{k} = \frac{2\pi}{0.8}$ $\lambda = 7.85 \text{ cm}$ $f = \frac{\omega}{2\pi} = \frac{16\pi}{2\pi}$ $f = 8 \text{ Hz}$ $v = f\lambda = 8(7.85)$ $v = 62.8 \text{ cm s}^{-1}$	G1 JU1
b) i)	$y = 2A = 2(6)$ $y = 12 \text{ m}$ $\omega = 2\pi \text{ rad s}^{-1}$ $k = 2\pi \text{ m}^{-1}$	G1 JU1 JU1 JU1
b) ii)	$y = 2A \cos kx \sin \omega t$ $y = 12 \cos 2\pi x \sin 2\pi t$ where y_1, y_2 and x are in meters, t in second	JU1
c) i)	$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{300}{14.4 \times 10^{-3}}}$ $v = 144.34 \text{ m s}^{-1}$	G1 JU1
c) ii)	$L = 3 \frac{\lambda}{2}$ $\lambda = \frac{2}{3} L = \frac{2}{3} (180 \times 10^{-2})$ $\lambda = 1.2 \text{ m @ } 120 \text{ cm}$	K1 G1 JU1
TOTAL		16 marks
2.a)	$n = 1.78 \times 10^{23}$ $e = 1.6 \times 10^{-19} \text{ C}$ $t = 1 \text{ H} = 60 \times 60 \text{ s} = 3600 \text{ s}$ $Q = ne$ $ne = It$	

	$I = \frac{ne}{t} = \frac{(1.78 \times 10^{23})(1.6 \times 10^{-19})}{3600}$ $\therefore I = 7.91 \text{ A}$	G1 JU1									
2. b)	$A = 60.0 \text{ mm}^2 = 60.0 \times 10^{-6} \text{ m}^2$ $R = 0.50 \Omega$ $\rho = 1.68 \times 10^{-8} \Omega \text{ m}, l = ? \quad \rho = \frac{RA}{l}$ $l = \frac{RA}{\rho}$ $l = \frac{(0.50)(60.0 \times 10^{-6})}{1.68 \times 10^{-8}}$ $\therefore l = 1.79 \times 10^3 \text{ m}$	G1 JU1									
2. c) i)	<table border="1"> <thead> <tr> <th>CIRCUIT (Simplify)</th><th>EFFECTIVE RESISTANCE (Calculated)</th><th>CONCEPT (Same)</th></tr> </thead> <tbody> <tr> <td></td><td> $R_{34} = R_3 + R_4$ $R_{34} = 5 + 3 = 8 \Omega$ </td><td> SERIES $I_{34} = I_3 = I_4$ </td></tr> <tr> <td></td><td> $R_{eff} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_{34}}}$ $R_{eff} = \frac{1}{\frac{1}{2} + \frac{1}{4} + \frac{1}{8}}$ $\therefore R_{eff} = 1.14 \Omega$ </td><td> PARALLEL $V_{34} = V_2 = V_1 = V_T$ </td></tr> </tbody> </table> $R_{eff} = ?$ $\therefore R_{eff} = 1.14 \Omega$	CIRCUIT (Simplify)	EFFECTIVE RESISTANCE (Calculated)	CONCEPT (Same)		$R_{34} = R_3 + R_4$ $R_{34} = 5 + 3 = 8 \Omega$	SERIES $I_{34} = I_3 = I_4$		$R_{eff} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_{34}}}$ $R_{eff} = \frac{1}{\frac{1}{2} + \frac{1}{4} + \frac{1}{8}}$ $\therefore R_{eff} = 1.14 \Omega$	PARALLEL $V_{34} = V_2 = V_1 = V_T$	K1 GJ1 K1 GJU1
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c) ii)	$V_T = I_T R_{eff}$ $I_T = \frac{V_T}{R_{eff}} = \frac{12}{1.14}$ $\therefore I_T = 10.53 \text{ A}$	G1 JU1									
c) iii)	$V_2 = V_T$ $\therefore V_2 = 12 \text{ V}$	K1 JU1									
TOTAL		12 marks									

3. a) i)	$C_1 \& C_2 \text{ series}$ $\frac{1}{C_{12}} = \frac{1}{6\mu} + \frac{1}{4\mu}$ $C_{12} = 2.4 \times 10^{-6} \text{ F}$ $C_3 \& C_4 \text{ series}$ $\frac{1}{C_{34}} = \frac{1}{12\mu} + \frac{1}{3\mu}$ $C_{34} = 2.4 \times 10^{-6} \text{ F}$ $C_{1234} = 2.4 \times 10^{-6} + 2.4 \times 10^{-6}$ $C_{1234} = 4.8 \times 10^{-6} \text{ F}$	G1 G1 GJU1
a) ii)	$V_{ab} = V_{12} = V_{34} = 12 \text{ V}$ $Q_{34} = Q_3 = Q_4$ $Q_3 = Q_{34} = C_{34}V_{34} = (2.4 \times 10^{-6})(12) = 2.88 \times 10^{-5} \text{ C}$	K1 K1 GJU1
b)		D2 D2
TOTAL		10 marks
4. a) i)	$Circumference = \frac{4}{10} = 2\pi r$ $r = 0.06 \text{ m}$	K1 J1
	$B = \frac{\mu_0 NI}{2r}$	

	$B = \frac{4\pi \times 10^{-7} (10)(20 \times 10^{-3})}{2(0.06)}$ $B = 2.09 \times 10^{-6} \text{ T}$	G1 JU1
a) ii)	Example:  *any suitable answer is accepted	D2 Direction of I (1 m) Direction of B (1 m)
b) i)	$B = \frac{\mu_0 NI}{L}$ $2.0 \times 10^{-3} = \frac{4\pi \times 10^{-7} (1200)(I)}{1.5}$ $I = 1.99 \text{ A}$	G1 JU1
b) ii)	$B = \frac{\mu_0 NI}{L}$ $B = \frac{4\pi \times 10^{-7} (1200)(1.99)}{1.8}$ $B = 1.67 \text{ mT}$	G1 JU1
c)	Change in magnetic flux $\Delta\phi = NBA \cos\theta - 0$ $= (150)(0.6)(5 \times 10^{-4})(1) - 0$ $= 0.045 \text{ Wb}$ Induced emf $\varepsilon = -\frac{d\phi}{dt}$ $\varepsilon = -\frac{0.045}{0.05}$ $= -0.9 \text{ V}$	G1 JU1 G1 JU1
TOTAL		14 marks
5. a) i)	$M = 2, \quad u = 2.5 \text{ cm}$ $M = -\frac{v}{u}$ $2 = -\frac{v}{2.5}$ $\therefore v = -5 \text{ cm}$	G1 JU1
a) ii)	Virtual	J1
a) iii)	$\frac{2}{r} = \frac{1}{u} + \frac{1}{v}$	G1

	$\frac{2}{r} = \frac{1}{2.5} + \frac{1}{-5}$ $\therefore r = +10 \text{ cm}$	JU1
a) iv)	 D2 (correct 2 rays with arrow – 1 mark) (correct image & label – 1 mark)	
b) i)	$v = -\frac{1}{3}u = -0.33u$ $f = -20 \text{ cm (f negative sign)}$ $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ $\frac{1}{-20} = \frac{1}{u} + \frac{1}{-0.33u}$ $u = 40.6 \text{ cm}$	K1 K1 G1 JU1
b) ii)	 Ray 1 Ray 2 any 2 rays - 1 mark Ray 3 Upright Image with label - 1 mark Object with label – 1 mark	D3
TOTAL		14 marks
6. a) i)	$m = 3$ $y_m = \frac{m\lambda D}{d}$ $y_3 = \frac{(3)(650 \times 10^{-9})(1.3)}{0.4 \times 10^{-3}}$ $y_3 = 6.34 \times 10^{-3} \text{ m}$	K1 G1 JU1

a) ii)	$m = 1$ $y_m = \frac{(m + \frac{1}{2})\lambda D}{d}$ $y_1 = \frac{(1 + \frac{1}{2})(650 \times 10^{-9})(1.3)}{0.4 \times 10^{-3}}$ $y_1 = 3.17 \times 10^{-3} \text{ m}$	K1 G1 JU1
b) i)	$\Delta y = \frac{\lambda D}{d}$ $\Delta y = \frac{(3.0 \times 10^{-7})(1.5)}{0.6 \times 10^{-6}}$ $\Delta y = 7.5 \times 10^{-4} \text{ m}$	G1 JU1
b) ii)	$\Delta y = \frac{\lambda D}{d}$ $\Delta y \propto D$ D increase, ΔY also increase so fringe separation become wider.	K1 J1
c)	3rd dark fringe; $m = 2$ 9th bright fringe; $m = 9$ $Y_{9,bright} - Y_{2,dark}$ $= \frac{9\lambda D}{d} - \frac{(2 + \frac{1}{2})\lambda D}{d}$ $= \frac{(6.5)(550 \times 10^{-9})(1.2)}{0.5 \times 10^{-3}}$ $= 8.58 \times 10^{-3} \text{ m}$	K1 K1 G1 JU1
TOTAL		14 marks
TOTAL		80